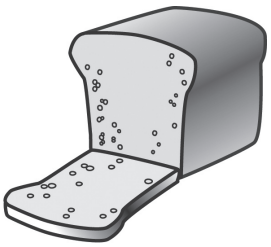
Fermentation

For hundreds of years people have used microorganisms such as yeast and bacteria to make foods and drinks such as beer, yoghurt, wine and cheese using a process called **fermentation**. Fermentation is the process where an organism converts a carbohydrate (starch or sugar) into alcohol or an acid.



☞ **1** Fill in the missing vowels (a, e, i, o, u) to complete the text below.

Bacteria

B __ ct __ ri __ are m __ cr __ __ rgan __ sms used in the process of f __ rm __ nt __ ti __ n to make yoghurt and ch __ __ se. The b __ ct __ r __ a feed on lactose, which is a s __ g __ r in milk, converting it to lactic __ c __d. The increased acidity thickens and s __ urs the m __ lk. The word equation for the fermentation of milk is:

$$\underset{\text{(sugar)}}{\text{lactose}} \xrightarrow{\text{bacteria}} \text{lactic acid}$$

Yeast

Y __ __ st is a single-celled microscopic f __ ng __ s that uses s __ g __ r as food. Yeast is used to make br __ __ d, b __ __ r and w __ n __. When bread is made, the y __ __ st feeds on sugar (gl __ c __ s __) in the bread d __ __ gh. The yeast is used to make b __ bbl __ s in the dough. As the dough starts to f __ rm __ nt, c __ rb __ n d __ __ x __ d __ is produced. The carbon dioxide is tr __ pp __ d in the dough, causing it to r __ s __. The w __ rd __ qu __ t __on for this is:

$$\underset{\text{(sugar)}}{\text{glucose}} \xrightarrow{\text{yeast}} \text{alcohol + carbon dioxide}$$

When the dough is baked, the yeast is k __ ll __ d and the alcohol __v __ por __ t __ s.

In the br __ w __ ng of beer, the alc __h __ l remains, along with the carbon dioxide, which gives b __ __ r its fizz.

2 What is the reactant in the process of fermentation used to make bread? _____

3 What are the products in the process of fermentation used to make bread?

_____

4 What product do bacteria produce during the process of fermentation? _____

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